

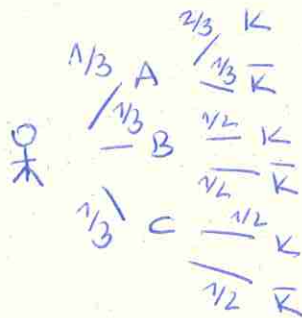
2019/07 (GITA MATE)

5. BLOKEA

- 3 TALDE: A, B, C

- KIKOLA $A(2/3)$, $B(1/2)$, $C(1/2)$

- KIROUK ET EGIN ETA A-KOA DELA?

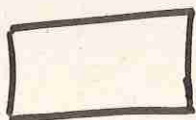


$$\underline{P(A/\bar{K})} = \frac{P(\bar{K} \cap A)}{P(\bar{K})} = \frac{1/9}{4/9} = \underline{1/4}$$

$$P(\bar{K} \cap A) = 1/3 \cdot 1/3 = 1/9$$

$$\begin{aligned} P(\bar{K}) &= 1/3 \cdot 1/3 + 1/3 \cdot 1/2 + 1/3 \cdot 1/2 = \frac{1}{9} + \frac{1}{6} + \frac{1}{6} = \frac{1}{9} + \frac{2}{6} = \\ &= \frac{1}{9} + \frac{1}{3} = \frac{1+3}{9} = \frac{4}{9} \end{aligned}$$

2018 EKAMA (GITA MATE)



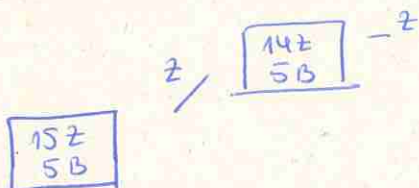
15 Z
5 B

a) Rusaz zuri

$$P(z) = \frac{15}{20} = \frac{3}{4}$$

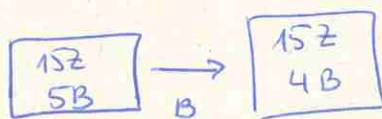
15 bola zuri gutira 20 bol izanik

b) Bi bola zuri.



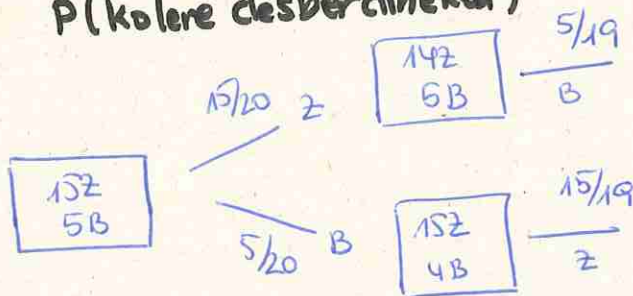
$$P(z \cap z) = \frac{15}{20} \cdot \frac{14}{19} = \frac{210}{380} = \frac{21}{38}$$

c) 1. belka 2. ena belka?



$$P(2. Belka) = \frac{4}{19}$$

d) $P(\text{kolore desberdinek})$



$$P(\text{Kolore desberdin}) = P(Z \cap B) + P(B \cap Z) = \frac{15}{20} \cdot \frac{5}{19} + \frac{5}{20} \cdot \frac{15}{19} =$$

$$= \frac{75}{380} + \frac{75}{380} = \frac{150}{380} = \frac{15}{38}$$

2016/07 (GITA MARE)

1000 ikasle

D. normala $\mu = 70$ $\sigma = 20$

a) $P(x > 80)$

c) %95 prob, kalikarao altueraz.

b) $P(x < 50)$

a) Banakete normal tipikatura horbelduko da.

$$P(x > 80) = P\left(z > \frac{80-70}{20}\right) = P\left(z > \frac{10}{20}\right) = P(z > 0.5) = \\ = 1 - P(z \leq 0.5) = \underline{0.3085}$$

$$b) P(x < 50) = P\left(z < \frac{50-70}{20}\right) = P\left(z < -\frac{20}{20}\right) = P(z < -1) = \\ = P(z > 1) = 1 - P(z < 1) = \underline{0.1587}$$

$$c) P(x < n) = P\left(z < \frac{n-70}{20}\right) = 0.95$$

$$\frac{n-70}{20} = 1.64 \quad n = 102.8 \quad \underline{n = 103 \text{ ikasle}}$$

PROBABILITATE BANAKETA BINOMIALA

LOTEKIA DAKTELEN 9011K SARIKEN BAT IBAZATI.

FAMILI BATEK 46 ZENBAKI ERUSI. PROBABILITATEA GUTXIENEZ 10 ZENBAKIK SARIA?

Banaketak binomialak, erakuste izateko probabilitatea 0.11 da $B(46, 0.11)$.

Banaketak normalen hurbilketako saiakera egongo da.

$$\mu = 46 \cdot 0.11 = 5.06 > 5 \quad \text{hurbilketak posible da}$$

$$\sigma = \sqrt{n \cdot p \cdot q} = \sqrt{46 \cdot 0.11 \cdot 0.89} = \sqrt{4.5034} = 2.12$$

$$B(46, 0.11) \sim N(5.06, 2.12) \sim N(0, 1)$$

$$P(X \geq 10) = P(X' \geq 9.5) = P\left(Z \geq \frac{9.5 - 5.06}{2.12}\right) =$$

$$= P\left(Z \geq \frac{4.44}{2.12}\right) =$$

$$= P(Z \geq 2.09) = 1 - P(Z \leq 2.09) =$$

$$= 1 - 0.9817 = 0.0183$$

